nearby or, at the very least, far away (such as the distant stars), that can serve as a reference for our various states of motion. In this example, however, there is absolutely no way for you to distinguish “not spinning” from “spinning” by comparisons with other material objects; there *aren’t* any other material objects. Mach took this observation to heart and extended it one giant step further. He suggested that in this case there might also be no way to *feel* a difference between various states of spinning. More precisely, Mach argued that in an otherwise empty universe there is *no distinction* between spinning and not spinning—there is no conception of motion or acceleration if there are no benchmarks for comparison—and so spinning and not spinning are the same. If Newton’s two rocks tied together by a rope were set spinning in an otherwise empty universe, Mach reasoned that the rope would remain slack. If you spun around in an otherwise empty universe, your arms and legs would not splay outward, and the fluid in your ears would be unaffected; you’d feel nothing.

This is a deep and subtle suggestion. To really absorb it, you need to put yourself into the example earnestly and fully imagine the black, uniform stillness of totally empty space. It’s not like a dark room in which you feel the floor under your feet or in which your eyes slowly adjust to the tiny amount of light seeping in from outside the door or window; instead, we are imagining that there are *no* things, so there is no floor and there is absolutely no light to adjust to. Regardless of where you reach or look, you feel and see absolutely nothing at all. You are engulfed in a cocoon of unvarying blackness, with no material benchmarks for comparison. And without such benchmarks, Mach argued, the very concepts of motion and acceleration cease to have meaning. It’s not just that you won’t feel anything if you spin; it’s more basic. In an otherwise empty universe, standing perfectly motionless and spinning uniformly are indistinguishable.[3]

Newton, of course, would have disagreed. He claimed that even completely empty space still has *space*. And, although space is not tangible or directly graspable, Newton argued that it still provides a something with respect to which material objects can be said to move. But remember how Newton came to this conclusion: He pondered rotating motion and *assumed* that the results familiar from the laboratory